Group B · Progression in year 2 Breast invasive carcinoma · TCGA PanCancer Atlas 2018 · model: neurosymbolic cox

1 · Patient and tumour record

Age at diagnosis	**78**	Tumour type	**Breast Invasive Ductal Carcinoma**
Sex	**Female**	OncoTree code	**IDC**
Race	**Asian**	PAM50 subtype	**Her2**
Ethnicity	**Not Hispanic Or Latino**	AJCC pathologic stage	**STAGE IIIA**
Genetic ancestry	**—**	Pathologic T	**T1**
Prior diagnosis	**No**	Pathologic N	**N2**
Radiation therapy	**No**	Pathologic M	**M0**
Neoadjuvant therapy	**No**	ICD-O-3 histology	**8500/3**

Sample: **TCGA-C8-A12Q-01** · Aneuploidy score: **13** · TMB (nonsyn., /Mb): **3.27** · MSIsensor score: **0.35** · AJCC edition: **—**

2 · Progression assessment

Model risk (cohort percentile)

97th

High risk tier · fused neuro-symbolic model



Probability of progression

27% by 36 mo

12 mo 8% · 24 mo 15% · 60 mo 42%

Model-estimated progression time

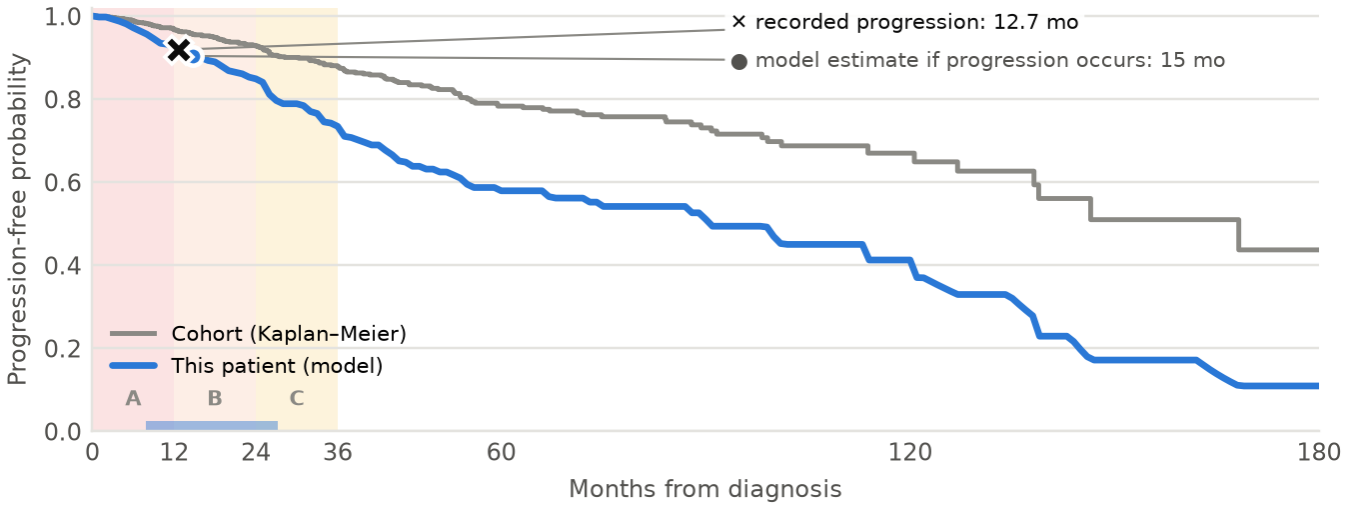
≈ 15 months

Window 12–24 months · IQR 8–27 mo if progression occurs

Recorded progression

12.7 months

Group B · Progression in year 2
✓ estimate in recorded window · |error| 2 mo



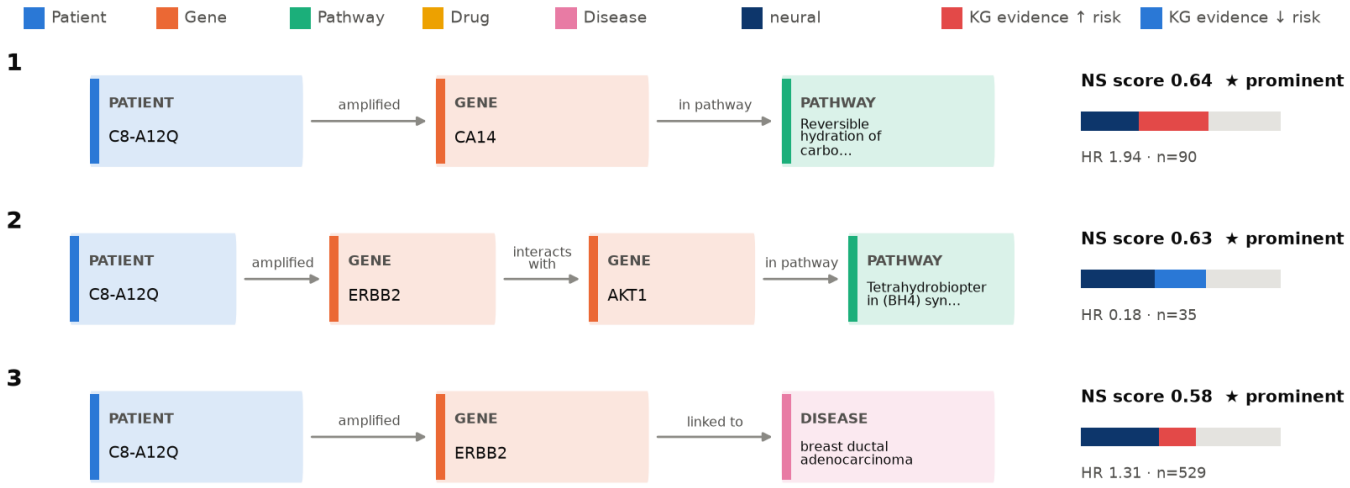
Blue: this patient's progression-free survival curve from the fused model, fitted on the four folds that exclude the patient. Grey: Kaplan–Meier curve of the whole cohort. Shaded bands mark the report groups A–C; the bar on the time axis is the model's interquartile range for the month of progression, given that progression occurs.

Summary

This 78-year-old patient has a Her2 tumour, AJCC stage III (T1, N2, M0). The fused neuro-symbolic model places them in the **97th risk percentile** (high tier). The most significant rule covering them (*M stage = M0 AND 40.5 ≤ Age < 86.5*) marks a group with a progression hazard 0.4× that of the remaining patients. The highest-scoring knowledge-graph path runs from **CA14** to **Reversible hydration of carbon dioxide**; in the training folds, carriers of this PrimeKG fact showed faster progression (HR 1.94). Progression was recorded at 12.7 months; the model's estimate given progression was 15 months.

3 · Neuro-symbolic reasoning

3a · Top knowledge-graph paths (GNN-scored, PrimeKG-grounded)

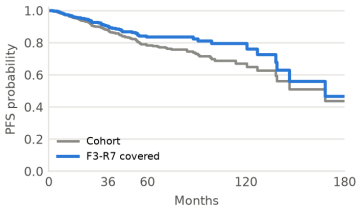


3 of 60 candidate paths shown; all clear the prominence cutoff (NS ≥ 0.57, the cohort's 80th percentile), capped at K = 5. NS score = 0.6 × neural percentile (GNNExplainer importance from the held-out fold model) + 0.4 × symbolic strength (|log-rank z| of the path's PrimeKG fact on the training folds, capped at 3).

#	Path	Neural pct	PrimeKG fact evidence (training folds)	NS
1 ★	**TCGA-C8-A12Q** amplified → **CA14** in pathway → **Reversible hydration of carbon dioxide**	48	↑ **faster progression** HR 1.94, z +2.6, n = 90 carriers of “Reversible hydration of carbon dioxide”	**0.64**
2 ★	**TCGA-C8-A12Q** amplified → **ERBB2** interacts with → **AKT1** in pathway → **Tetrahydrobiopterin (BH4) synthesis, recycling, salvage and regulation**	62	↓ **slower progression** HR 0.18, z -1.9, n = 35 carriers of “Tetrahydrobiopterin (BH4) synthesis, recycling, salvage and regulation”	**0.63**
3 ★	**TCGA-C8-A12Q** amplified → **ERBB2** linked to → **breast ductal adenocarcinoma**	65	↑ **faster progression** HR 1.31, z +1.4, n = 529 carriers of “breast ductal adenocarcinoma”	**0.58**

3b · Symbolic survival rules that fire for this patient (RuleKit LR-Rules)

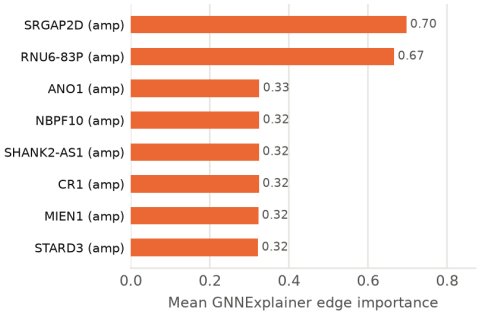
Rule	IF …	Covered	HR vs rest	Median PFS
F3-R7 ↓ **lower risk**	M stage = M0 AND 40.5 ≤ Age < 86.5	635 pts 66 events	**0.37** p 1.1e-07	168 mo



Rules were learned by RuleKit's log-rank rule induction on the four folds that exclude this patient; counts, hazard ratios, and Kaplan–Meier medians are on that training data. [KG] marks a rule whose body contains a PrimeKG fact.

4 · Molecular profile and knowledge-graph context

Gene	Alteration	Detail	PrimeKG links	GNN imp.
CCND1 · driver	High-level amplification	GISTIC ±2	26	0.32
KMT2C · driver	Mutation	p.I323=	4	0.31
ERBB2 · driver	High-level amplification	GISTIC ±2	55	0.31
TP53 · driver	Mutation	p.Y205N	57	0.31
SRGAP2D	High-level amplification	GISTIC ±2	0	0.70
RNU6-83P	High-level amplification	GISTIC ±2	0	0.67
ANO1	High-level amplification	GISTIC ±2	2	0.33
NBPF10	High-level amplification	GISTIC ±2	0	0.32
SHANK2-AS1	High-level amplification	GISTIC ±2	0	0.32
CR1	High-level amplification	GISTIC ±2	3	0.32
MIEN1	High-level amplification	GISTIC ±2	0	0.32
STARD3	High-level amplification	GISTIC ±2	1	0.32



67 altered panel genes in total; known drivers first, then by GNN importance. CNA follows the graph policy: only homozygous deletion (–2) and high-level amplification (+2).

PrimeKG drug–protein links of the altered genes

CCND1 → Acetylsalicylic acid (DB00945); CCND1 → Arsenic trioxide (DB01169); CCND1 → Encorafenib (DB11718); CCND1 → Bryostatin 1 (DB11752); ERBB2 → Trastuzumab (DB00072); ERBB2 → Lapatinib (DB01259); ERBB2 → IGN311 (DB04988); ERBB2 → Trastuzumab emtansine (DB05773); ERBB2 → Varlitinib (DB05944); ERBB2 → AV-412 (DB06021) … and 55 more. These are knowledge-graph links (target, enzyme, carrier, or transporter), shown for research context only; they are not treatment recommendations.

Case-based context

Most similar patients	Age	Stage	Subtype	Similarity	Recorded outcome
TCGA-BH-A0HY	60	I	LumB	0.71	no progression, followed 50.8 mo
TCGA-C8-A138	54	III	Her2	0.67	no progression, followed 12.5 mo
TCGA-A8-A07I	69	III	Her2	0.66	no progression, followed 14.0 mo
TCGA-B6-A0I9	62	IV	Her2	0.66	progressed at 11.9 mo (group A)
TCGA-AR-A254	50	III	Her2	0.66	no progression, followed 85.6 mo

Cosine similarity over stage, subtype, histology, age, driver alterations, and PrimeKG pathway facts. Outcomes are recorded follow-up, shown as context, not as a prediction.

How to read this report

The risk and the survival curve come from a Cox model on the clinical block (age, sex, PAM50 subtype, AJCC stage, M stage) plus the GNN risk and the RuleKit rule risk, all fitted on the four folds that exclude this patient. In cross-validation the GNN and rule terms carry close to zero weight once the clinical block is present, so the risk mostly reflects clinical stage, nodal status, age, and subtype. The knowledge-graph paths and rules explain which PrimeKG facts the graph model attended to and how patients sharing those facts fared; they are evidence, not causal claims. The progression-time estimate (loglogistic aft on progressors) is within 12 months of the recorded month for 51% of held-out progressors and in the recorded window for 34%; the recorded month is always shown beside it.